Math Thermo
class # 23
4/21/2026

Application: Two-Phase Incompressible Flow

We will assume that

1) $r = 2$ (binary mixture),

2) $\Psi_i \equiv 0$, $i = 1, 2$ (no external potential),

3) $\sigma_k \equiv 0$, $k = 1, \dots, K$ (no reactions),

4) $\rho \equiv \rho_0 \equiv$ constant (incompressible).

Since the mixture is incompressible, the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$$

is replaced by

$$\nabla \cdot \vec{v} = 0$$

The partial densities are

$$\rho_i = \rho_0 \phi_i, \quad i = 1, 2.$$

where

$$\phi_1 + \phi_2 = 1$$

are the mass fractions. We will assume that

$$U = \int_\Omega \hat{u}(T, \phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2)\, d\vec{x}$$

$$F = \int_\Omega \hat{f}(T, \phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2)\, d\vec{x}$$

$$S = \int_\Omega \hat{s}(T, \phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2)\, d\vec{x}$$

To shorten the writing, we will use the notation

$$\hat{u}(T, \phi, \nabla\phi) = \hat{u}(T, \phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2).$$

We will assume (more on this later) that

$$\hat{f} = \hat{u} - T\hat{s} \tag{23.1}$$

as in equilibrium. This is another consequence of local thermodynamic equilibrium.

Further, we assume, for simplicity,

$$\rho u =: \hat{u}(T, \phi, \nabla\phi) = \tilde{u}(T) + \lambda_u \delta(\phi, \nabla\phi),$$

$$\rho f =: \hat{f}(T, \phi, \nabla\phi) = \tilde{f}(T) + \lambda_f \delta(\phi, \nabla\phi),$$

$$\rho s =: \hat{s}(T, \phi, \nabla\phi) = \tilde{s}(T) + \lambda_s \delta(\phi, \nabla\phi),$$

where λ_u and λ_s are constants.

In other words, the T and ϕ dependencies are separable. For $\tilde{u}, \tilde{f}, \tilde{s}$ we will eventually take

$$\tilde{u} = \rho_0 C_h T,$$

$$\tilde{f} = \rho_0 C_h T - \rho_0 C_h T \ln\left(\frac{T}{T_0}\right),$$

$$\tilde{s} = \rho_0 C_h \ln\left(\frac{T}{T_0}\right).$$

Clearly,

$$\tilde{f} = \tilde{u} - T\tilde{s}.$$

For the gradient term, we use the model

$$\delta(\phi, \nabla\phi) = \frac{1}{\varepsilon}\frac{\phi_1^2\phi_2^2}{4} + \frac{\varepsilon}{2}\left(\frac{|\nabla\phi_1|^2}{2} + \frac{|\nabla\phi_2|^2}{2}\right),$$

where $\varepsilon > 0$ has units of length. $[\delta] = \text{length}^{-1}$.

Finally, we note that, it must be true that

$$\lambda_f = \lambda_u - T\lambda_s.$$

Now, let us note that

$$\frac{\partial\tilde{s}}{\partial T} = \rho_0 C_h \frac{1}{T}$$

$$\frac{\partial\tilde{u}}{\partial T} = \rho_0 C_h$$

So that

(23.2) $$\boxed{\frac{\partial\tilde{s}}{\partial T} = \frac{1}{T}\frac{\partial\tilde{u}}{\partial T}.}$$

This condition is called <u>thermal compatibility</u>. We will generally assume this holds.

Now, we are going to derive the entropy production terms in a slightly diffent way. Rather than relying on a Gibbs relation, we will directly calculate the RHS of

$$\frac{\partial \hat{s}}{\partial t} + \nabla\cdot\left(\hat{s}\vec{v}\right) = -\nabla\cdot\vec{J}_S + \sigma_S$$

in order to determine J_S^D and σ_S.

Later on we will explain the rationale for this proceedure. Before we do this, here are the governing equations:

(23.3) $$\nabla\cdot\vec{v} = 0,$$

(23.4) $$\rho_0\frac{\partial\phi_i}{\partial t} + \rho_0\nabla\cdot\left(\phi_i\vec{v}\right) = -\nabla\cdot\vec{J}_i^D, \qquad (i=1,2)$$

(23.5) $$\rho_0\frac{\partial\vec{v}}{\partial t} + \rho_0\nabla\cdot\left(\vec{v}\otimes\vec{v}\right) + \nabla P = -\nabla\cdot\Pi,$$

(23.6) $$\frac{\partial\hat{u}}{\partial t} + \nabla\cdot\left(\hat{u}\vec{v}\right) = -\nabla\cdot\vec{J}_U^D - \Pi:\nabla\vec{v},$$

where we have used $\Psi_i \equiv 0$, $\tilde{\sigma}_K \equiv 0$, $\nabla\cdot\vec{v}\equiv 0$, $\rho_0 \equiv$ const, et cetera.

We must find Π, $\vec{J}_i^D$, $\vec{J}_U^D$ so that entropy is always increasing. We need some preliminary calculations.

<u>Proposition</u> (23.1): Let us assume that

$$\rho u =: \hat{u} = \tilde{u}(T) + \lambda_u\,\delta(\phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2).$$

Then, if $\vec{v}$ is the barycentric velocity, we have

$$\frac{\partial\hat{u}}{\partial t} + \nabla\cdot\left(\hat{u}\vec{v}\right) = \frac{\partial\tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t} + \nabla\cdot\left(T\vec{v}\right)\right)$$

$$+ \lambda_u \omega_2 \left(\frac{\partial \phi_2}{\partial t} + \nabla \cdot (\phi_2 \vec{v}) \right)$$

$$+ \lambda_u \omega_1 \left(\frac{\partial \phi_1}{\partial t} + \nabla \cdot (\phi_1 \vec{v}) \right)$$

(23.7) $$+ \frac{\lambda_u}{2} \varepsilon \nabla \cdot \left(\nabla \phi_2 \frac{\partial \phi_2}{\partial t} \right)$$

$$+ \frac{\lambda_u}{2} \varepsilon \nabla \cdot \left(\nabla \phi_1 \frac{\partial \phi_1}{\partial t} \right)$$

$$+ \frac{\lambda_u}{2} \varepsilon \left[\nabla \cdot (\nabla \phi_2 \otimes \nabla \phi_2) \right] \cdot \vec{v}$$

$$+ \frac{\lambda_u}{2} \varepsilon \left[\nabla \cdot (\nabla \phi_1 \otimes \nabla \phi_1) \right] \cdot \vec{v},$$

where

$$\omega_i = \frac{1}{\varepsilon} \frac{\partial W}{\partial \phi_i} - \frac{\varepsilon}{2} \Delta \phi_i, \qquad i = 1, 2,$$

and

$$W(\phi_1, \phi_2) = \frac{1}{4} \phi_1^2 \phi_2^2 .$$

Proof: We use the general form

$$\rho u = \hat{u} = \tilde{u}(T) + \lambda_u \left(\frac{1}{\varepsilon} W(\phi_1, \phi_2) + \frac{\varepsilon}{2} \left(\frac{|\nabla \phi_1|^2}{2} + \frac{|\nabla \phi_2|^2}{2} \right) \right).$$

Now,

$$\frac{\partial u}{\partial t} = \frac{\partial \tilde{u}}{\partial T} \frac{\partial T}{\partial t} + \lambda_u \left(\frac{1}{\varepsilon} \frac{\partial W}{\partial \phi_1} \frac{\partial \phi_1}{\partial t} + \frac{\partial W}{\partial \phi_2} \frac{\partial \phi_2}{\partial t} \right.$$

$$\left. + \frac{\varepsilon}{2} \nabla \phi_1 \cdot \nabla \left(\frac{\partial \phi_1}{\partial t} \right) + \frac{\varepsilon}{2} \nabla \phi_2 \cdot \nabla \left(\frac{\partial \phi_2}{\partial t} \right) \right)$$

Next,

$$\nabla \cdot (\hat{u} \vec{v}) = \nabla \cdot (\tilde{u}(T) \vec{v}) + \frac{\lambda_u}{\varepsilon} \nabla \cdot (W(\phi_1, \phi_2) \vec{v})$$

$$+ \frac{\lambda_u \varepsilon}{4} \nabla \cdot (|\nabla \phi_1|^2 \vec{v}) + \frac{\lambda_u \varepsilon}{4} \nabla \cdot (|\nabla \phi_2|^2 \vec{v}).$$

Notice that,

$$\nabla\cdot(\tilde{u}(T)\vec{v}) = \nabla(\tilde{u}(T))\cdot\vec{v} + \tilde{u}(T)(\cancelto{0}{\nabla\cdot\vec{v}})$$

$$= \frac{\partial\tilde{u}}{\partial T}\nabla T\cdot\vec{v}$$

$$= \frac{\partial\tilde{u}}{\partial T}\nabla\cdot(T\vec{v}).$$

Likewise,

$$\nabla\cdot(W(\phi_1,\phi_2)\vec{v}) = \nabla(W(\phi_1,\phi_2))\cdot\vec{v} + W(\phi_1,\phi_2)(\cancelto{0}{\nabla\cdot\vec{v}})$$

$$= \frac{\partial W}{\partial\phi_1}\nabla\phi_1\cdot\vec{v} + \frac{\partial W}{\partial\phi_2}\nabla\phi_2\cdot\vec{v}$$

$$= \frac{\partial W}{\partial\phi_1}\nabla\cdot(\phi_1\vec{v}) + \frac{\partial W}{\partial\phi_2}\nabla\cdot(\phi_2\vec{v})$$

Finally, consider the following calculation: using Einstein's summation convention,

$$\nabla\cdot(|\nabla\phi|^2\vec{v}) = \frac{\partial}{\partial x_i}\left(\frac{\partial\phi}{\partial x_j}\frac{\partial\phi}{\partial x_j}v_i\right)$$

$$= \frac{\partial^2\phi}{\partial x_i\partial x_j}\frac{\partial\phi}{\partial x_j}v_i + \frac{\partial\phi}{\partial x_j}\frac{\partial^2\phi}{\partial x_i\partial x_j}v_i + \frac{\partial\phi}{\partial x_j}\frac{\partial\phi}{\partial x_j}\cancelto{0}{\frac{\partial v_i}{\partial x_i}}$$

$$= 2\frac{\partial\phi}{\partial x_j}\frac{\partial^2\phi}{\partial x_i\partial x_j}v_i.$$

But note that

$$[\nabla\cdot(\nabla\phi\otimes\nabla\phi)]\cdot\vec{v} = \frac{\partial}{\partial x_j}\left(\frac{\partial\phi}{\partial x_i}\frac{\partial\phi}{\partial x_j}\right)v_i$$

$$= \frac{\partial^2\phi}{\partial x_i\partial x_j}\frac{\partial\phi}{\partial x_j}v_i + \frac{\partial\phi}{\partial x_i}v_i\frac{\partial^2\phi}{\partial x_j\partial x_j}$$

$$= \frac{1}{2}\nabla\cdot(|\nabla\phi|^2\vec{v}) + \vec{v}\cdot\nabla\phi\,\Delta\phi.$$

Thus,

$$\nabla\cdot\left(|\nabla\phi|^2\vec{v}\right) = 2\left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v}$$
$$- 2\,\vec{v}\cdot\nabla\phi\,\Delta\phi$$

Putting everything together,

$$\begin{aligned}
\frac{\partial u}{\partial t} + \nabla\cdot\left(u\vec{v}\right) = {} & \frac{\partial\tilde{u}}{\partial T}\frac{\partial T}{\partial t} + \lambda_u\left(\frac{1}{\varepsilon}\frac{\partial W}{\partial\phi_1}\frac{\partial\phi_1}{\partial t} + \frac{\partial W}{\partial\phi_2}\frac{\partial\phi_2}{\partial t}\right. \\
& \left. + \frac{\varepsilon}{2}\nabla\phi_1\cdot\nabla\left(\frac{\partial\phi_1}{\partial t}\right) + \frac{\varepsilon}{2}\nabla\phi_2\cdot\nabla\left(\frac{\partial\phi_2}{\partial t}\right)\right) \\
& + \frac{\partial\tilde{u}}{\partial T}\nabla\cdot\left(T\vec{v}\right) \\
& + \frac{\lambda_u}{\varepsilon}\left(\frac{\partial W}{\partial\phi_1}\nabla\cdot\left(\phi_1\vec{v}\right) + \frac{\partial W}{\partial\phi_2}\nabla\cdot\left(\phi_2\vec{v}\right)\right) \\
& + \frac{\lambda_u\varepsilon}{2}\left[\nabla\cdot\left(\nabla\phi_1\otimes\nabla\phi_1\right)\right]\cdot\vec{v} \\
& + \frac{\lambda_u\varepsilon}{2}\left[\nabla\cdot\left(\nabla\phi_2\otimes\nabla\phi_2\right)\right]\cdot\vec{v} \\
& - \frac{\lambda_u\varepsilon}{2}\Delta\phi_1\,\nabla\cdot\left(\phi_1\vec{v}\right) \\
& - \frac{\lambda_u\varepsilon}{2}\Delta\phi_2\,\nabla\cdot\left(\phi_2\vec{v}\right).
\end{aligned}$$

We use the identity

$$\nabla\phi\cdot\nabla\left(\frac{\partial\phi}{\partial t}\right) = \nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right) - \Delta\phi\,\frac{\partial\phi}{\partial t}$$

and some simplification to obtain

$$
\begin{aligned}
\frac{\partial u}{\partial t} + \nabla\cdot\left(u\vec{v}\right) = {} & \frac{\partial \tilde{u}}{\partial T}\frac{\partial T}{\partial t} + \lambda_u\left(\frac{1}{\varepsilon}\frac{\partial W}{\partial \phi_1}\frac{\partial \phi_1}{\partial t} + \frac{\partial W}{\partial \phi_2}\frac{\partial \phi_2}{\partial t}\right. \\
& + \frac{\varepsilon}{2}\nabla\cdot\left(\nabla\phi_1\frac{\partial \phi_1}{\partial t}\right) - \frac{\varepsilon}{2}\Delta\phi_1\frac{\partial \phi_1}{\partial t} \\
& \left. + \frac{\varepsilon}{2}\nabla\cdot\left(\nabla\phi_2\frac{\partial \phi_2}{\partial t}\right) - \frac{\varepsilon}{2}\Delta\phi_2\frac{\partial \phi_2}{\partial t}\right) \\
& + \frac{\partial \tilde{u}}{\partial T}\nabla\cdot\left(T\vec{v}\right) \\
& + \frac{\lambda_u}{\varepsilon}\left(\frac{\partial W}{\partial \phi_1}\nabla\cdot\left(\phi_1\vec{v}\right) + \frac{\partial W}{\partial \phi_2}\nabla\cdot\left(\phi_2\vec{v}\right)\right) \\
& + \frac{\lambda_u\varepsilon}{2}\left[\nabla\cdot\left(\nabla\phi_1\otimes\nabla\phi_1\right)\right]\cdot\vec{v} \\
& + \frac{\lambda_u\varepsilon}{2}\left[\nabla\cdot\left(\nabla\phi_2\otimes\nabla\phi_2\right)\right]\cdot\vec{v} \\
& - \frac{\lambda_u\varepsilon}{2}\Delta\phi_1\nabla\cdot\left(\phi_1\vec{v}\right) \\
& - \frac{\lambda_u\varepsilon}{2}\Delta\phi_2\nabla\cdot\left(\phi_2\vec{v}\right) \\
= {} & \frac{\partial \tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t} + \nabla\cdot\left(T\vec{v}\right)\right) \\
& + \lambda_u\omega_1\left(\frac{\partial \phi_1}{\partial t} + \nabla\cdot\left(\phi_1\vec{v}\right)\right) \\
& + \lambda_u\omega_2\left(\frac{\partial \phi_2}{\partial t} + \nabla\cdot\left(\phi_2\vec{v}\right)\right) \\
& + \frac{\lambda_u\varepsilon}{2}\nabla\cdot\left(\nabla\phi_1\frac{\partial \phi_1}{\partial t}\right) \\
& + \frac{\lambda_u\varepsilon}{2}\nabla\cdot\left(\nabla\phi_2\frac{\partial \phi_2}{\partial t}\right) \\
& + \frac{\lambda_u\varepsilon}{2}\left[\nabla\cdot\left(\nabla\phi_1\otimes\nabla\phi_1\right)\right]\cdot\vec{v} \\
& + \frac{\lambda_u\varepsilon}{2}\left[\nabla\cdot\left(\nabla\phi_2\otimes\nabla\phi_2\right)\right]\cdot\vec{v} \quad ///
\end{aligned}
$$

Proposition (23.2): Recall that, since $r=2$,

$$\phi_1+\phi_2=1.$$

As a consequence, setting

$$\phi:=\phi_2,$$

it follows that

(23.8)
$$\begin{aligned}\frac{\partial \hat{u}}{\partial t}+\nabla\cdot(\hat{u}\vec{v}) &= \frac{\partial \tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t}+\nabla\cdot(T\vec{v})\right)\\ &+\lambda_u\omega\left(\frac{\partial\phi}{\partial t}+\nabla\cdot(\phi\vec{v})\right)\\ &+\lambda_u\varepsilon\nabla\cdot\left(\nabla\phi\frac{\partial\phi}{\partial t}\right)\\ &+\lambda_u\varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v},\end{aligned}$$

where

$$\omega:=\frac{1}{\varepsilon}\tilde{W}'(\phi)-\varepsilon\Delta\phi,\qquad \tilde{W}(\phi):=W(1-\phi,\phi).$$

Proof: Clearly,

$$\frac{\partial\phi_1}{\partial t}+\nabla\cdot(\phi_1\vec{v}) = -\frac{\partial\phi}{\partial t}-\nabla\cdot(\phi\vec{v})$$

$$\frac{\partial\phi_2}{\partial t}+\nabla\cdot(\phi_2\vec{v}) = \frac{\partial\phi}{\partial t}+\nabla\cdot(\phi\vec{v})$$

$$\begin{aligned}I &:= \lambda_u\omega_1\left(\frac{\partial\phi_1}{\partial t}+\nabla\cdot(\phi_1\vec{v})\right)+\lambda_u\omega_2\left(\frac{\partial\phi_2}{\partial t}+\nabla\cdot(\phi_2\vec{v})\right)\\ &= \lambda_u(\omega_1-\omega_2)\left(\frac{\partial\phi}{\partial t}-\nabla\cdot(\phi\vec{v})\right)\\ &= \lambda_u\left(\frac{1}{\varepsilon}\frac{\partial W}{\partial\phi_1}-\frac{1}{\varepsilon}\frac{\partial W}{\partial\phi_2}-\frac{\varepsilon}{2}\Delta\phi_1+\frac{\varepsilon}{2}\Delta\phi_2\right)\times\end{aligned}$$

$$\left(\frac{\partial \phi}{\partial t} - \nabla\cdot(\phi\vec{v})\right)\Big)$$

$$= \lambda_u \left(\frac{1}{\varepsilon}\frac{\partial \tilde{W}}{\partial \phi} - \varepsilon\Delta\phi\right) \times \left(\frac{\partial \phi}{\partial t} - \nabla\cdot(\phi\vec{v})\right)\Big)$$

where

$$\tilde{W}(\phi) := W(1-\phi, \phi).$$

The rest of the details are left for an exercise. ///

The entropy equation will have the form

$$\frac{\partial \hat{s}}{\partial t} + \nabla\cdot(\hat{s}\vec{v}) = -\nabla\cdot J_s^D + \sigma_s.$$

We calculated the terms of this equation previously using the Gibbs relation

$$ds = \frac{1}{T}du + \frac{P}{T}dv - \sum_{i=1}^{r}\frac{\mu_i}{T}d\phi_i,$$

where s is the specific entropy, u is the specific internal energy density, v is the specific volume.

This equation might not hold any longer, since

$$S_v(t) = \int_{V(t)} \hat{s}(T, \phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2)\, d\vec{x}$$

rather than

$$S_v(t) = \int_{V(t)} \rho\, s(u, v, \phi_1, \phi_2)\, d\vec{x}.$$

In the present case, we have a more straight -

forward way of finding J_s^D and σ_s.

Proposition (23.3): Suppose that thermal compatibility holds, ie,

(23.9) $$\frac{\partial \tilde{s}}{\partial T} = \frac{1}{T}\frac{\partial \tilde{u}}{\partial T}$$

and assume that

(23.10) $$\rho u =: \hat{u}(T, \phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2) = \tilde{u}(T) + \lambda_u \delta(\phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2),$$

(23.11) $$\rho f =: \hat{f}(T, \phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2) = \tilde{f}(T) + \lambda_f \delta(\phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2),$$

(23.12) $$\rho s =: \hat{s}(T, \phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2) = \tilde{s}(T) + \lambda_s \delta(\phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2),$$

with

$$\tilde{f} = \tilde{u} - T\tilde{s} \quad \text{and} \quad \lambda_f = \lambda_u - T\lambda_s.$$

Then,

(23.13) $$\frac{\partial \hat{s}}{\partial t} + \nabla\cdot(\hat{s}\vec{v}) = -\nabla\cdot\vec{J}_s^D + \sigma_s,$$

where

(23.14) $$\vec{J}_s^D = \frac{1}{T}J_v^D - \frac{\lambda_f}{\rho_0}\frac{\omega}{T}\vec{J}_2^D + \lambda_f\varepsilon\frac{1}{T}\nabla\phi\frac{\partial\phi}{\partial t} + \lambda_f\varepsilon\frac{1}{T}(\nabla\phi\otimes\nabla\phi)\vec{v}$$

(23.15) $$\sigma_s = -\frac{1}{T}\nabla\vec{v} : \left(\Pi - \lambda_f\varepsilon\nabla\phi\otimes\nabla\phi\right) - \frac{\lambda_f}{\rho_0}\nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^D + \nabla\left(\frac{1}{T}\right)\cdot\left\{J_v^D + \lambda_f\varepsilon\nabla\phi\frac{\partial\phi}{\partial t} + \lambda_f\varepsilon(\nabla\phi\otimes\nabla\phi)\vec{v}\right\}$$

Proof: Using what we learned from Proposition (23.2), we must have

$$\frac{\partial \hat{s}}{\partial t} + \nabla\cdot(\hat{s}\vec{v}) = \frac{\partial \tilde{s}}{\partial T}\left(\frac{\partial T}{\partial t} + \nabla\cdot(T\vec{v})\right) + \lambda_s \omega\left(\frac{\partial \phi}{\partial t} + \nabla\cdot(\phi\vec{v})\right) + \lambda_s \varepsilon \nabla\cdot\left(\nabla\phi \frac{\partial \phi}{\partial t}\right) + \lambda_s \varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v}, \tag{23.16}$$

where

$$\omega := \frac{1}{\varepsilon}\tilde{W}'(\phi) - \varepsilon\Delta\phi, \qquad \tilde{W}(\phi) := W(1-\phi,\phi).$$

Recall, we have identified

$$\phi := \phi_2 \quad (\phi_1 = 1-\phi).$$

Using thermal compatibility

$$\frac{\partial \hat{s}}{\partial t} + \nabla\cdot(\hat{s}\vec{v}) = \frac{1}{T}\frac{\partial \tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t} + \nabla\cdot(T\vec{v})\right) + \lambda_s \omega\left(\frac{\partial \phi}{\partial t} + \nabla\cdot(\phi\vec{v})\right) + \lambda_s \varepsilon \nabla\cdot\left(\nabla\phi \frac{\partial \phi}{\partial t}\right) + \lambda_s \varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v},$$

or, equivalently,

$$T\left(\frac{\partial \hat{s}}{\partial t}+\nabla\cdot(\hat{s}\vec{v})\right) = \frac{\partial \tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t}+\nabla\cdot(T\vec{v})\right) + T\lambda_s\,\omega\left(\frac{\partial \phi}{\partial t}+\nabla\cdot(\phi\vec{v})\right) + T\lambda_s\varepsilon\,\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right) + T\lambda_s\varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v}. \tag{23.17}$$

Using

$$\frac{\partial \hat{u}}{\partial t} + \nabla\cdot(\hat{u}\vec{v}) = -\nabla\cdot\vec{J}_U^{\,0} - \Pi : \nabla\vec{v} \tag{23.6}$$

and

$$\frac{\partial \hat{u}}{\partial t}+\nabla\cdot(\hat{u}\vec{v}) = \frac{\partial \tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t}+\nabla\cdot(T\vec{v})\right) + \lambda_u\,\omega\left(\frac{\partial \phi}{\partial t}+\nabla\cdot(\phi\vec{v})\right) + \lambda_u\varepsilon\,\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right) + \lambda_u\varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v} \tag{23.8}$$

we have our <u>heat equation</u>:

$$\frac{\partial \tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t}+\nabla\cdot(T\vec{v})\right) = -\nabla\cdot\vec{J}_U^{\,0} - \Pi : \nabla\vec{v} - \lambda_u\,\omega\left(\frac{\partial \phi}{\partial t}+\nabla\cdot(\phi\vec{v})\right) - \lambda_u\varepsilon\,\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right) - \lambda_u\varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v}. \tag{23.18}$$

Substituting (23.18) into (23.17) we have

$$
\begin{aligned}
T\left(\frac{\partial \hat{s}}{\partial t} + \nabla\cdot\left(\hat{s}\vec{v}\right)\right) &= -\nabla\cdot\vec{J}_U^D - \Pi : \nabla\vec{v} \\
&\quad + \left(T\lambda_s - \lambda_u\right)\omega\left(\frac{\partial \phi}{\partial t} + \nabla\cdot(\phi\vec{v})\right) \\
&\quad + \left(T\lambda_s - \lambda_u\right)\varepsilon\nabla\cdot\left(\nabla\phi\,\frac{\partial \phi}{\partial t}\right) \\
&\quad + \left(T\lambda_s - \lambda_u\right)\varepsilon\left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v} \\
&= -\nabla\cdot\vec{J}_U^D - \Pi : \nabla\vec{v} \\
&\quad - \lambda_f\,\omega\left(\frac{\partial \phi}{\partial t} + \nabla\cdot(\phi\vec{v})\right) \\
&\quad - \lambda_f\,\varepsilon\nabla\cdot\left(\nabla\phi\,\frac{\partial \phi}{\partial t}\right) \\
&\quad - \lambda_f\,\varepsilon\left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v}.
\end{aligned}
$$

Using

(23.4) $$\rho_0\frac{\partial \phi_i}{\partial t} + \rho_0\nabla\cdot\left(\phi_i\vec{v}\right) = -\nabla\cdot\vec{J}_i^D$$

we have

$$
\begin{aligned}
T\left(\frac{\partial \hat{s}}{\partial t} + \nabla\cdot\left(\hat{s}\vec{v}\right)\right) &= -\nabla\cdot\vec{J}_U^D - \Pi : \nabla\vec{v} \\
&\quad + \lambda_f\,\omega\,\frac{\nabla\cdot\vec{J}_2^D}{\rho_0} \\
&\quad - \lambda_f\,\varepsilon\nabla\cdot\left(\nabla\phi\,\frac{\partial \phi}{\partial t}\right) \\
&\quad - \lambda_f\,\varepsilon\left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v}.
\end{aligned}
$$

Equivalently,

(23.19)
$$\frac{\partial \hat{s}}{\partial t} + \nabla\cdot\left(\hat{s}\vec{v}\right) = \frac{-\nabla\cdot\vec{J}_U^D}{T} - \frac{\Pi : \nabla\vec{v}}{T} + \frac{\lambda_f\,\omega}{\rho_0 T}\nabla\cdot\vec{J}_2^D - \frac{\lambda_f\,\varepsilon}{T}\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right) - \frac{\lambda_f\,\varepsilon}{T}\left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v}.$$

Now, we use some identities:

$$\frac{\omega}{T}\nabla\cdot\vec{J}_2^D = \nabla\cdot\left(\frac{\omega}{T}\vec{J}_2^D\right) - \nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^D,$$

$$\frac{1}{T}\nabla\cdot\vec{J}_U^D = \nabla\cdot\left(\frac{1}{T}\vec{J}_U^D\right) - \nabla\left(\frac{1}{T}\right)\cdot\vec{J}_U^D,$$

$$\frac{1}{T}\nabla\cdot\left(\nabla\phi\frac{\partial\phi}{\partial t}\right) = \nabla\cdot\left(\frac{1}{T}\nabla\phi\frac{\partial\phi}{\partial t}\right) - \nabla\left(\frac{1}{T}\right)\cdot\nabla\phi\frac{\partial\phi}{\partial t}.$$

$$\left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v} = \nabla\cdot\left[\left(\nabla\phi\otimes\nabla\phi\right)\vec{v}\right] - \left(\nabla\phi\otimes\nabla\phi\right):\nabla\vec{v}$$

Inserting into (23.19), we find

(23.20)
$$\frac{\partial \hat{s}}{\partial t} + \nabla\cdot\left(\hat{s}\vec{v}\right) = -\nabla\cdot\left(\frac{1}{T}\vec{J}_U^D\right) + \nabla\left(\frac{1}{T}\right)\cdot\vec{J}_U^D + \frac{\lambda_f}{\rho_0}\nabla\cdot\left(\frac{\omega}{T}\vec{J}_2^D\right) - \frac{\lambda_f}{\rho_0}\nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^D - \lambda_f\varepsilon\nabla\cdot\left(\frac{1}{T}\nabla\phi\frac{\partial\phi}{\partial t}\right) + \lambda_f\varepsilon\nabla\left(\frac{1}{T}\right)\cdot\nabla\phi\frac{\partial\phi}{\partial t} - \frac{\lambda_f\,\varepsilon}{T}\nabla\cdot\left[\left(\nabla\phi\otimes\nabla\phi\right)\vec{v}\right]$$

$$+\frac{\lambda_f \varepsilon}{T}(\nabla\phi\otimes\nabla\phi):\nabla\vec{v} \qquad \frac{-\Pi:\nabla\vec{v}}{T}$$

One more identity should do it:

$$\frac{1}{T}\nabla\cdot\left[(\nabla\phi\otimes\nabla\phi)\vec{v}\right] = \nabla\cdot\left[\frac{1}{T}(\nabla\phi\otimes\nabla\phi)\vec{v}\right] - \nabla\left(\frac{1}{T}\right)\cdot\left[(\nabla\phi\otimes\nabla\phi)\vec{v}\right].$$

Then (23.20) becomes

(23.21)
$$\begin{aligned}\frac{\partial\hat{s}}{\partial t}+\nabla\cdot(\hat{s}\vec{v}) = & -\nabla\cdot\left(\frac{1}{T}\vec{J}_v^D\right) + \nabla\left(\frac{1}{T}\right)\cdot\vec{J}_v^D \\ & + \frac{\lambda_f}{\rho_0}\nabla\cdot\left(\frac{\omega}{T}\vec{J}_2^D\right) - \frac{\lambda_f}{\rho_0}\nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^D \\ & - \lambda_f\varepsilon\nabla\cdot\left(\frac{1}{T}\nabla\phi\frac{\partial\phi}{\partial t}\right) + \lambda_f\varepsilon\nabla\left(\frac{1}{T}\right)\cdot\nabla\phi\frac{\partial\phi}{\partial t} \\ & + \frac{\lambda_f\varepsilon}{T}(\nabla\phi\otimes\nabla\phi):\nabla\vec{v} \qquad \frac{-\Pi:\nabla\vec{v}}{T} \\ & - \lambda_f\varepsilon\nabla\cdot\left[\frac{1}{T}(\nabla\phi\otimes\nabla\phi)\vec{v}\right] \\ & + \lambda_f\varepsilon\nabla\left(\frac{1}{T}\right)\cdot\left[(\nabla\phi\otimes\nabla\phi)\vec{v}\right].\end{aligned}$$

Comparing (23.21) to (23.13), we find

$$\begin{aligned}J_s^D = & \frac{1}{T}J_v^D - \frac{\lambda_f}{\rho_0}\frac{\omega}{T}\vec{J}_2^D + \lambda_f\varepsilon\frac{1}{T}\nabla\phi\frac{\partial\phi}{\partial t} \\ & + \lambda_f\varepsilon\frac{1}{T}(\nabla\phi\otimes\nabla\phi)\vec{v}\end{aligned}$$

$$\begin{aligned}\sigma_s = & -\frac{1}{T}\nabla\vec{v}:\left(\Pi - \lambda_f\varepsilon\nabla\phi\otimes\nabla\phi\right) - \frac{\lambda_f}{\rho_0}\nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^D \\ & + \nabla\left(\frac{1}{T}\right)\cdot\left\{J_v^D + \lambda_f\varepsilon\nabla\phi\frac{\partial\phi}{\partial t} + \lambda_f\varepsilon(\nabla\phi\otimes\nabla\phi)\vec{v}\right\}\end{aligned}$$

Thus (23.14) and (23.15) are confirmed. ///

Other forms are possible. Consider the following identity:

$$\nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^{\,D} = \frac{1}{T}\nabla\omega\cdot\vec{J}_2^{\,D} + \nabla\left(\frac{1}{T}\right)\cdot\omega\vec{J}_2^{\,D}.$$

Then,

(23.22) $$\sigma_s = -\frac{1}{T}\nabla\vec{v}:\left(\Pi - \lambda_f\varepsilon\nabla\phi\otimes\nabla\phi\right) - \frac{\lambda_f}{\rho_0 T}\nabla\omega\cdot\vec{J}_2^{\,D} + \nabla\left(\frac{1}{T}\right)\cdot\left\{ J_U^{\,D} + \lambda_f\varepsilon\nabla\phi\frac{\partial\phi}{\partial t} + \lambda_f\varepsilon(\nabla\phi\otimes\nabla\phi)\vec{v} - \frac{\lambda_f}{\rho_0}\omega\vec{J}_2^{\,D}\right\}.$$